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RV COLLEGE OF ENGINEERING®
 (An Autonomous Institution Affiliated to VTU)
 IV Semester B. E. Examinations Sept/Oct – 2024
 Aerospace Engineering

AEROSPACE PROPULSION

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Any missing data may be suitably assumed.

PART-A

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1	1.1	Calculate the isentropic efficiency of the compressor if air is compressed adiabatically from 15°C and 1.03 bar to 182°C and 4 times the pressure.	02	3	2
	1.2	Name the cooling technique used to ensure even temperature distribution across a turbine blade.	01	2	1
	1.3	Determine the thermal efficiency of an ideal Brayton Cycle if air is compressed from 1 bar and 300 K to 6 bars and the maximum temperature is 1100 K with heat supplied being 100 MW.	02	3	2
	1.4	How does the propulsion efficiency of a jet engine change with increasing thrust power and increasing mass flow rate?	01	2	4
	1.5	In which component of a turbojet engine does maximum pressure loss typically occur?	01	1	3
	1.6	Find the fuel flow rate if a jet engine produces 6000 N of thrust with a specific thrust of 580 Ns/kg and a fuel-to-air ratio of 0.0198.	02	3	2
	1.7	Calculate the thrust produced by a propeller with a diameter of 2.5m, flying at 500 km/h at an altitude of 8000 m, given an air density of 0.525 kg/m ³ and a flight-to-jet speed ratio of 0.7.	02	3	1
	1.8	If a solid propellant rocket motor has a positive initial temperature sensitivity of burn rate, what parameter must be increased to achieve greater thrust and reduced burn time?	01	2	2
	1.9	Determine the thrust produced by a solid rocket motor with an end-burning grain of cross-sectional area 0.4 m ² , a propellant density of 1500 kg/m ³ , a linear regression rate of 5mm/s, and specific impulse of 200s.	02	1	3
	1.10	In a turbojet engine, the primary function of the afterburner is to _____ by burning additional fuel in the exhaust stream.	01	1	4
	1.11	The component in a jet engine responsible for converting the thermal energy of the exhaust gases into mechanical energy is the _____.	01	1	1
	1.12	The principle behind the operation of a turbojet engine is based on the _____.	01	1	3
	1.13	For a gas turbine engine operating on the Brayton cycle, if the pressure ratio is 8 and the temperature at the turbine inlet is 1200 K, the maximum temperature in the cycle, assuming an isentropic expansion and a compressor efficiency of 85%, is approximately _____ K.	02	3	4
	1.14	In solid rocket propulsion, the term _____ refers to the amount of thrust produced per unit mass flow rate of propellant.	01	2	2

PART-B

2	a	Why thrust augmentation is necessary for pure turbojet engines. Justify the reasons enumerating the methods adopted for augmentation. Illustrate the working of an afterburner technique with appropriate diagrams.	08	1	3
	b	An aircraft manufacturer is choosing between a turboprop and a turbofan engine for a regional aircraft intended to operate on a route with the following conditions: - Cruising speed: 741 km/h - Cruising Altitude: 6,096 meters (20,000 feet) - Flight Distance: 1,482 km - Fuel Cost: Rs 71.1 per liter - Passenger Load: 50 passengers - Specific Fuel Consumption (SFC): - Turboprop: 0.30 kg/kW/h - Turbofan: 0.61 kg/kN/h - Thrust Required: - Turboprop: 2,237 kW - Turbofan: 22.24 kN - Fuel Density: 0.8 kg/liter Choose a suitable engine for the flight operation and justify the reason for choosing	08	3	1
3	a	Define the following performance parameters and give their mathematical expressions. - Specific Thrust - Specific Impulse - Thrust power - Thermal Efficiency	08	2	2
	b	A turboprop engine operates at an altitude of 3 Km above mean sea level and an aircraft speed of 525 kmph. The data for the engine is as given below: Compressor efficiency = 0.79 Velocity of air at compressor entry = 90 m/s Temperature rise through compressor = 230°C Estimate the following for the engine i) Compressor pressure ratio ii) Power required by compressor per unit flow rate of air iii) Air standard efficiency of the engine	08	3	3
		OR			
4	a	An aircraft flies at 960kmph. One of its turbojet engines takes 40kg/s of air and expands the gases to ambient pressure. The air fuel ratio is 50 and lower calorific value is 43MJ/kg. for maximum thrust power determine the following: - Jet Velocity - Thrust Developed - Specific Thrust - Thrust power - TSFC - Propulsive, thermal and overall efficiencies	10	3	4
	b	Derive the generalized thrust equation for a turbojet engine and discuss the importance of terms involved in the expression.	06	1	1

5	a	Illustrate the working principle of a convergent nozzle and discuss the effect of back pressure on the mass flow rate of the nozzle	10	2	2
	b	You are supposed to design a small gas turbine engine intended for a UAV (Unmanned Aerial Vehicle) application. The engine must operate efficiently at varying speeds and provide sufficient thrust for the UAV's mission profile. The design requirements are: - Pressure Ratio: The engine needs a pressure ratio of approximately 4:1. - Flow Rate: The engine needs to handle a mass flow rate of 5 kg/s. - Engine Size: The engine should be compact and lightweight to fit within the UAV's constraints. - Efficiency: The engine must have high efficiency to maximize fuel economy. Based on the design requirements, would an axial compressor or a centrifugal compressor be more suitable for this small turbine engine? Explain your choice with proper justification	06	5	3
OR					
6	a	Illustrate the characteristic features of Supersonic inlets and its various modes of operation	08	2	3
	b	Discuss in detail about the typical axial turbine stage and how it contributes to the energy conversion process	08	1	1
7	a	A solid rocket engine is to be designed for a small satellite launch. The rocket needs to have: High fuel efficiency (specific impulse), Stable and predictable thrust during the flight, A compact and lightweight design. Which type of progressive burning will you decide to be best for this rocket engine? Justify why you think this choice would be the most suitable based on the needs listed above.	08	4	4
	b	How do you decide the proportions for a typical composite solid propellant. What are the functions of its individual constituents?	08	1	3
OR					
8	a	Illustrate the working of a liquid propellant rocket engine highlighting its advantages and disadvantages.	10	2	3
	b	Enumerate the critical requirements while selecting propellants for liquid propellant rockets.	06	2	4
9	a	Compare pump and gas pressure-fed liquid rocket propulsion system	10	2	4
	b	Define and give the mathematical expressions for the following rocket performance parameters: - Characteristic Jet Velocity - Propellant Mass Fraction - Impulse-to-weight Ratio	06	2	3
OR					

10 a	A rocket projectile has the following characteristics - Initial Mass = 200kg - Mass after rocket operation = 130 kg - Payload, non-propulsive structure etc = 110 kg - Rocket operation duration = 3.0 seconds - Average specific impulse = 240 seconds - Determine the vehicle and propellant mass ratio, propellant mass fraction, propellant flow rate, thrust, thrust-to-weight ratio, effective exhaust velocity, total impulse, and the impulse to weight ratio.	08	4	4
b	Illustrate the working principle of ion thruster and nuclear fission thruster rocket propulsion system.	08	2	3